

Response to Reviewers

Manuscript (revised): *Do Cryptocurrency Markets Differentiate Infrastructure from Regulatory Shocks? A Multi-Moment Event Study with Dependence-Robust Inference* **Original title:** *Infrastructure vs Regulatory Shocks: Asymmetric Volatility Response in Cryptocurrency Markets* **Journal:** *Digital Finance* (Springer) **Author:** Murad Farzulla (Dissensus) **Decision:** Major Revision

Overview: this revision overturns its own headline, on purpose

I want to set out the shape of this revision before the point-by-point response, because it is larger than the decision letter requested and it changes the paper's central claim.

The original submission's headline was that infrastructure events generate a **5.7× larger conditional-variance response** than regulatory events, with $p < 0.001$. (That 5.7× is the prior paper's high-salience-curated estimate under its canonical estimator and rolling winsorisation; the present standalone pipeline does not reproduce it and instead gives a reproducible curated point estimate of **4.88×**, which is the consistent headline figure throughout this revision — the estimator bookkeeping is set out under R2.1 and in F8 of the pre-submission-audit section.) Reviewer 2's central comment (R2.1) asked whether that multiplier survives when the event-selection screen is relaxed. Answering it properly required me to reconstruct the candidate pool and re-run the inference end to end — and that process surfaced two defects in the inference behind the headline. Neither is a data problem; neither changes the point estimate. Both are errors in how significance was computed, and once corrected, **the headline does not survive**. The asymmetry is directional but not statistically distinguishable from zero under dependence-robust inference.

Rather than patch the original claim, I have rebuilt the paper around the correction. The revision does four things:

1. **Reports a dual null at both moments, honestly.** I test the infrastructure-regulatory difference at the **first moment (returns)** and the **second moment (conditional variance)** on one shared sample under one matched dependence-robust design. Returns: +7.19 pp, block-bootstrap $p = 0.283$, Ibragimov–Müller $p = 0.295$. Variance: directional $\approx 2.6\text{--}5\times$, Student- t -copula bootstrap $p = 0.322$ (baseline), 0.318 (crisis control) — one-sided throughout this letter, matching the inference ladder; the manuscript's results table also reports the two-sided values (0.323, 0.321). Neither survives.
2. **Makes the inference cautionary tale the lead contribution.** A new section (§*Why the Asymmetry Was an Artefact*) sets out four inference lessons — pseudoreplication, the concentrating nature of cross-asset dependence, heavy-tailed bootstrap innovations, and block-versus-parametric resampling at the first moment — as a self-correction. This is framed as methodological maturity, not self-flagellation: I followed the correct inference and it dissolved my own result. The section is now anchored by two new pieces of apparatus: an **inference ladder** (a single table walking the *identical* comparison through seven progressively-more-correct inference procedures, from the dead headline $t = 4.768 / p = 0.0008$ down to the inference of record $p = 0.322$; the rungs *bracket* the comparison rather than climbing monotonically — the naive test far below at 0.0008, the low-power model-free proxies above near 0.5, and the calibrated t -copula bootstrap settling it at 0.322), and a **Monte-Carlo size study** (described below) that shows the over-rejection is a general property of the naive procedure, not a feature of this one dataset.
3. **Subsumes the companion returns analysis.** The first-moment (CAR) analysis previously circulated as a public preprint (*Same Returns, Different Risks*, arXiv:2602.07046), which is not under review

elsewhere, is folded into this manuscript, so the two moments are tested and reported together rather than across two papers. The overlap of the first-moment text with that preprint is disclosed here plainly.

4. **Keeps the two findings that survive** as ballast: the scope condition (curated vs mechanical event identification, R2.1) and the weekly Granger result (R2.3).

What I am explicitly *not* claiming: any significant asymmetry at either moment; that the effects are zero or the shock types identical (the first moment is underpowered — MDE $\approx$ 19–20 pp vs observed $\approx$ 7 pp — so this is a failure to reject); and a 5%-significance claim anywhere in the paper.

Disclosure: the two inference defects behind the original $p = 0.0008$ / $5.7\times$

I flag these directly so the reviewers do not have to reconstruct them by comparing rounds.

Defect 1 — pseudoreplication. The original significance test ($t = 4.768$, $p = 0.0008$) treated the six per-asset GJR-GARCH-X coefficients as six independent observations. They are not: all six assets see the *same* 50 events and correlate 0.54–0.83 (mean $\bar{\rho} \approx 0.69$), so the effective sample size is far below six. The coefficients cluster tightly because they average the same correlated events, not because the effect is precisely estimated. Event-level re-tests (the unit of independence) collapse it: Welch $p = 0.52$, Mann-Whitney $p = 0.54$, cluster-robust panel (SE clustered by event) $p = 0.50$. The model-free proxies are low-powered and bracket the truth from below; the uncorrected test brackets it from far above. A design-effect correction on the actual estimator — computed on the *contrast*’s cross-asset correlation ($\rho_d = 0.23$, the correlation of the per-asset infrastructure–regulatory *differences*, not the level correlation $\bar{\rho} = 0.69$), which gives $df_{eff} \approx 2.3$ and an effective $N \approx 2.8$ — lands the honest value at $t \approx 1.8$ – 2.3 , **$p \approx 0.07$ – 0.15** — marginal, never the 5% the uncorrected test implied. (The level correlation $\bar{\rho} = 0.69$ yields a smaller effective $N \approx 1.4$, but that is the penalty for *averaging the six coefficients*, not for the contrast we actually test; the manuscript’s Lesson-1 footnote derives the contrast value, and this letter now quotes it consistently.)

Defect 2 — Gaussian bootstrap innovations under heavy tails. Moving to a cross-asset CCC-GARCH-X bootstrap addressed the dependence but left a second defect: it drew standardised innovations from a **Gaussian** while each model is fitted with **Student- t** errors at $v \approx 3$. With $v \approx 3$ the true innovation is far heavier-tailed, so the Gaussian draw mis-specifies the null distribution and biases the bootstrap p downward. Correcting the draw to the fitted Student- t margins (a multivariate- t copula, per-asset v , rescaled to unit variance) moves the baseline p from **0.057 (Gaussian) to 0.322 (Student- t)**. The Gaussian value is the heavy-tail-naïve figure the t -copula corrects; it is never co-reported as a headline.

Together, the two corrections convert the original $p \approx 0.001$ into a failure to reject, with the point estimate ($\approx 5\times$) unchanged at every stage. Each correction is implemented in replication code (c5/c6 pseudoreplication re-tests; c7 Gaussian CCC bootstrap; c9 the corrected Student- t -copula bootstrap that is the inference of record).

A Monte-Carlo size study certifies the diagnosis (new since the prior round)

A reviewer could reasonably ask whether I have simply walked one dataset from a low p to a high one and called it a correction. To pre-empt that, I added a Monte-Carlo **size-distortion study** (c10) that turns the diagnosis from an assertion into a measured fact. I fit a true-null DGP to the data — per-asset GJR-GARCH-X estimated under a null-imposed *combined* event dummy (so $\delta_{infra} = \delta_{reg}$ by construction; there is genuinely no differential effect in the truth), cross-asset dependence at $\bar{\rho}_z = 0.705$, Student- t margins at the fitted v (median 3.82) with joint tail dependence via a true multivariate- t copula — then simulate 1,000 panels (923 usable) and measure how often each inference procedure rejects a true null.

The result is exactly the demonstration the paper needs:

Inference procedure	Size @ 0.05	Size @ 0.10
(i) Naive i.i.d. t -test	0.512 $\pm$ 0.016	0.701 $\pm$ 0.015
(ii) Design-effect, $t(N-1=5)$ reference	0.434 $\pm$ 0.016	0.655 $\pm$ 0.016
(ii') same rule, correct $t(df_eff \approx 2.3)$ reference	0.220	0.535
(iii) Gaussian-copula bootstrap	0.308 $\pm$ 0.015	0.398 $\pm$ 0.016
(iv) Student- t -copula bootstrap	0.039 $\pm$ 0.006	0.095 $\pm$ 0.010

The naive i.i.d. rule — the one that produced the headline $t = 4.768 / p = 0.0008$ — rejects a true null **51% of the time at the nominal 5% level** ($\approx 10\times$ nominal) and 70% at 10%. The design-effect rule improves sharply once its reference distribution is corrected for the effective degrees of freedom — from 43% under the wrong $t(N-1=5)$ tail to 22% under the correct $t(df_eff \approx 2.3)$ — but the analytic correction alone does not reach nominal size; the Gaussian-copula bootstrap (right dependence, wrong tails) improves but stays over-sized at 31%; only the Student- t -copula bootstrap restores nominal size (3.9% / 9.5%). Crucially, the t -copula rung passes a **self-consistency check**: it is calibrated from the same DGP the panels are drawn from, so its size *must* land near nominal up to Monte-Carlo error — and it does, which certifies the simulation has no bug. This certifies the implementation, not correctness on the real, unknown DGP; the t -copula's claim to be the inference of record rests instead on its getting both the dependence *and* the heavy tails right, which the alternatives demonstrably do not — and the conclusion needs only that *some* dependence-respecting, heavy-tailed procedure kills the significance, not that this one is uniquely perfect. The headline $p = 0.0008$ was, with high probability, one of the 51% of true nulls the naive rule falsely rejects. This study is new since the prior round and is reported as §A *Monte-Carlo size study* with its full DGP in an appendix.

Reviewer 1

“Overall, this is a strong and carefully executed manuscript that fits well within the scope of *Digital Finance*.”

The reviewer's comments largely concerned tone, framing, and presentation. Reviewer 1 was steering toward softer causal language while keeping the finding; the revision goes further than that only because correct inference required it, so R1's tone concern is now moot at the level of the conclusion itself — the claim that prompted the moderation request no longer stands. I address each point on its own terms below.

R1.1 — Tone and strength of causal language

Comment: “Some statements adopt very strong causal or normative language. A slight moderation would improve credibility.”

Response: This is now resolved at the level of the conclusion, not just the wording. The paper no longer asserts a significant asymmetry to soften — it reports a dual failure to reject. Every causal and significance claim attached to the headline has been removed or downgraded. The abstract, introduction, results, discussion, and conclusion state the result as “we cannot reject equality of infrastructure and regulatory event impacts at either moment, on this sample, under correct inference,” with the underpowered caveat stated symmetrically throughout. There is no 5%-significance claim anywhere in the manuscript.

R1.2 — Event classification and endogeneity (exogeneity framing)

Comment: *“Explicitly framing infrastructure events as closer to exogenous shocks would strengthen identification.”*

Response: Retained and still relevant. Infrastructure failures (unannounced outages, externally-discovered exploits, banking-partner insolvencies) arrive at calendar times essentially exogenous to current market conditions; regulatory timing is closer to endogenous (political cycles, prior market behaviour, leaks). The manuscript states this asymmetric identification quality explicitly. Note that under the new framing this strengthens the honesty of the null rather than the strength of a claim: even granting infrastructure events their cleaner exogeneity, the difference is not statistically resolvable on this sample.

R1.3 — Length and focus

Comment: *“Some robustness material could be moved to an online appendix to improve readability.”*

Response: The revised manuscript is reorganised around the dual-null result and the four inference lessons, which are now the spine of the paper rather than appended robustness. Secondary robustness (event-window and anticipation sweeps, winsorisation invariance, temporal/sub-sample stability, the structural-break control battery, and the AR(1)- and event-in-mean controls) is reported in compact form with headline numbers in the body and full per-asset tables in the replication materials. I am happy to relocate any block the editor prefers to an online appendix — the Monte-Carlo size-study DGP and the full per-asset parameter tables are the natural candidates.

R1.4 — Network analysis as a secondary contribution

Comment: *“The ETH versus BTC centrality result ... should be framed more clearly as a complementary finding.”*

Response: Resolved by removal. In rebuilding the paper around the dual-null result and the inference contribution, the network/centrality material was cut entirely — it is no longer in the manuscript. For the record, the reason it did not survive the refocus is that eigenvector centrality on a six-node correlation graph is weakly discriminating (ETH 1.00 vs BTC 0.97; all six between 0.85 and 1.00) and carried no causal or systemic-hub claim, so it added little against the cost in length and focus that Reviewer 1 also flagged (R1.3).

R1 Minor — Overlapping event windows

Comment: *“Clarify handling of overlapping event windows.”*

Response: The overlap-handling protocol is now stated explicitly in §3.1 (“Overlapping windows”). Because events can fall within days of one another, their windows overlap in calendar time; we handle this by construction rather than by dropping the affected events. For the variance-side dummies, same-type overlaps take the *union* (a day in any window is one, never double-counted) and cross-type overlaps leave *both* indicators active so the GJR-GARCH-X estimates the two contributions jointly; for the first moment, each event’s CAR is computed independently and aggregated at the event level, so overlaps do not double-count, and the dependence they induce between temporally close events is exactly what the event-level block bootstrap is built to preserve. This matters more under the new framing, since the shared 50-event sample now drives both moments and the treatment is therefore consistent across them.

R1 Minor — Near-integrated volatility persistence

Comment: “Briefly discuss near-integrated volatility persistence.”

Response: Addressed jointly with R2.2. Bai-Perron-style (PELT) break tests detect 2–4 breaks per asset clustering around the November 2022 FTX collapse; within-sub-sample persistence is on average lower than the full-sample $\alpha + \beta \approx 0.988$ (sub-sample mean ≈ 0.90), though within the wide estimation noise of the short regimes. The persistence drop is reported as suggestive, not significant (no sub-segment drop exceeds two standard errors) — in keeping with the paper’s general discipline of not over-reading.

R1 Minor — Positioning within the digital finance literature

Comment: “Strengthen positioning within the digital finance literature.”

Response: The literature section now positions the contribution against operational-resilience and microstructure work and, more centrally, against the methodological literature on pseudoreplication, few-cluster inference (Ibragimov–Müller), and heavy-tailed bootstrap calibration — which is where the paper’s lead contribution now sits.

Reviewer 2

“The economic significance of the findings is well-established... the author implements several robustness checks.”

Reviewer 2’s R2.1 (selection bias) is the comment that set this entire revision in motion. It receives the fullest response.

R2.1 — Selection bias from the “demonstrable market-wide impact” criterion

Comment: (Summarised) *The Stage-2 criterion (≥ 2 assets showing market-wide impact) risks selecting on the dependent variable; infrastructure failures almost always clear it while regulatory news may be systematically rejected, biasing the comparison toward H1. The reviewer asks for (a) a drop-out rate by event category and (b) a sensitivity analysis at a relaxed threshold.*

Response: This comment is the proximate cause of the whole reckoning, and it is answered on its own terms — by *measuring* selection bias rather than arguing it away — and then carried further than the reviewer asked, into the inference correction.

(a) Drop-out census. I reconstructed a systematic candidate pool of **135 events** (82 infrastructure, 53 regulatory, Jan 2019–Aug 2025) from public-record sources (Rekt/DeFiLlama hack leaderboards; SEC/CFTC enforcement records; major-jurisdiction regulatory announcements; exchange status pages) and applied the *identical* Stage-2 impact filter to both legs. Pass rates:

Category	N	Stage-2 pass rate
Infrastructure	82	64.6%
Regulatory	53	75.5%

The filter is **less** demanding for regulatory candidates than for infrastructure candidates (two-proportion $z = -1.33$, n.s.) — the *opposite* sign to the selection-bias hypothesis. The screen does not mechanically favour

infrastructure. Because the same filter is applied to both source-defined populations, the comparison inherits no asymmetric selection from how the pool was assembled.

Direct evidence that curation tracked event-type and salience, not realised volatility. The sharpest refutation of “selection on the dependent variable” is the *mismatch* between the curation criterion and the mechanical impact screen, which disagree in both directions. Of the 50 curated events, **14 fail the ≥ 2 -asset mechanical impact screen yet were retained** — 4 of them with *zero* assets moving beyond 1 sample-SD (scheduled regulatory milestones and the like). Conversely, **57 filter-passing candidates were dropped** from the curated set, **29 of which moved ≥ 5 of the 6 assets**. A curated sample that selected on realised volatility could produce neither pattern: it admits low-impact events for their type and salience and omits high-impact ones, which is the opposite of selecting on the outcome. (On the candidate pool itself: the original manual working list was not preserved as a structured artefact, so an earlier draft’s **208-candidate figure has been removed as unverifiable**; the reconstructed **135-candidate pool** is the screening pool used for this sensitivity analysis, disclosed transparently as a fresh reconstruction rather than the lost original.)

(b) Relaxed-threshold sensitivity — the scope condition. Re-estimating the GJR-GARCH-X specification under a properly-nested mechanical sweep on the reconstructed pool:

Spec	N infra	N reg	Multiplier	Welch p
Curated sample (full 4-stage protocol)	26	24	4.88× (headline)	0.0015
No filter (135)	82	53	0.58×	0.446
≥ 1 asset (115)	69	46	1.49×	0.318
≥ 2 assets (93, like-for-like)	53	40	1.61×	0.164
≥ 3 assets (77)	42	35	1.31×	0.403

On the broad pool the multiplier is a modest **1.3–1.6×**, **not significant at any threshold**; the like-for-like two-asset screen (the mechanical analogue of the curated impact criterion) gives 1.61×, nowhere near the curated $\approx 5\times$. The strong magnitude is therefore a property of **curated, high-salience event identification**, not of a mechanical impact threshold. The $\sim 3\times$ curated-vs-mechanical gap holds under *both* winsorisation rules — 4.88× vs 1.61× under the headline global-clip winsorisation, and 5.24× vs 1.63× under rolling-bound winsorisation — so it is an identification effect, not a winsorisation artefact.

A note on the three curated figures, so they are never conflated. The consistent headline throughout is **4.88×** (global-clip winsorisation, canonical estimator). The **5.24×** reported as a winsorisation-robustness check is the *FastTARCHX* estimator under rolling winsorisation. The prior paper’s **5.7×** is the *canonical* estimator under that *same* rolling winsorisation ($\delta_{\text{infra}}/\delta_{\text{reg}} = 2.385/0.419$). The 5.24-vs-5.7 gap is therefore an **estimator difference** — concentrated entirely in δ_{reg} (0.464 for FastTARCHX vs 0.419 canonical) — **not a winsorisation difference**; and the 5.7× is not reproduced by the present standalone pipeline. The manuscript states this explicitly (§Specification stability).

Correction relative to the prior revision draft (disclosed). An intermediate draft mis-framed the sweep as “peaking at the two-asset threshold,” placing the curated 50-event sample at the “two-asset point.” That was wrong on two counts: the curated sample is selected by the full four-stage protocol, not a mechanical two-asset screen, and its count (50) is smaller than the three-asset screen (77), so the counts cannot nest. The genuine two-asset screen on the reconstructed pool (93 events) gives 1.61× ($p = 0.16$), and the result is now presented as a scope condition, not a peak.

How R2.1 escalated into the inference correction. Re-running the estimation to answer R2.1 is what exposed that the curated-sample significance itself (not just the magnitude) did not survive proper inference. The curated-sample significance behind the $\approx 5\times$ point estimate (the apparent $p < 0.001$) was a pseudoreplication-plus-Gaussian-bootstrap artefact (see Overview, Defects 1–2). So the selection-bias answer has two layers: (i) the magnitude is identification-dependent (scope condition), and (ii) even the curated-sample significance evaporates under dependence-robust inference (t -copula $p = 0.322$). Both are reported. The scope condition is retained as a substantive finding; it is no longer load-bearing for a significance claim, because there is none.

R2.2 — Bai-Perron structural-break tests

Comment: “*The high persistence parameter often masks structural breaks ... perform formal tests for multiple structural changes (Bai and Perron, 1998).*”

Response: Implemented and retained, and expanded into a full control battery. A PELT-based multiple-break procedure (BIC-penalised, trimming 0.15, max 5 breaks, bootstrap CIs on break locations) detects 2–4 breaks per asset, clustering at the November 2022 FTX collapse; within-sub-sample persistence falls from a full-sample mean of $\alpha + \beta \approx 0.985$ to ≈ 0.905 . I am now explicit that this drop is **descriptive only** — when each sub-segment’s persistence is estimated with its own standard error, *none* of the 27 sub-segments falls more than two SEs below its full-sample value (the regimes are short, SEs ± 0.06 – 0.31), and the drop is not a small-sample artefact (mean persistence 0.891 for $n \geq 500$ segments vs 0.887 for $n < 500$). The near-integration framing is therefore qualified, not overturned, and reported as suggestive.

The break regimes also serve the second-moment inference directly, and the revision now reports a fuller break-control battery with its **cross-asset-robust** significance (the prior round’s break-control p -values were naive Welch tests — the same pseudoreplication the headline suffered). Under the copula bootstrap: adding a single FTX-crisis-regime control moves the multiplier only from $4.88\times$ to $3.97\times$ and the robust p from 0.322 to 0.318 (a real chunk of the magnitude is crisis-baseline variance, but the differential is not *merely* the crisis regime); the full-regime control halves the multiplier to $2.64\times$ ($p = 0.249$, reliability-flagged at 23% degenerate refits) and inverts XRP alone. The complete battery — break controls, anticipation sweep, winsorisation invariance, Ljung-Box/AR(1), and persistence SEs — is in §*Control battery for the second-moment point estimate*. A formal Markov regime-switching estimate of crisis amplification (H3) is motivated by the break evidence but left as future work; given the dual null it is moot for the headline and I prefer not to report an untested amplification.

Anticipation, disclosed as a new robustness exhibit. I also added an anticipation sweep: if regulatory events were anticipated, lengthening the regulatory pre-window would absorb leaked pre-event volatility into δ_{reg} and shrink the multiplier toward one. The opposite occurs — as the reg pre-window lengthens from 3 to 10 days, δ_{reg} *falls* ($0.405 \rightarrow 0.224$), the infrastructure coefficient is invariant, and the multiplier *grows* ($4.88\times \rightarrow 8.77\times$). Anticipation runs against the asymmetry, not for it. This is defensive ammunition for the point estimate, not a significance claim (which remains the copula bootstrap’s, and null).

R2.3 — Granger causality for sentiment and volatility

Comment: “*The relationship between news sentiment and volatility could be reflexive ... consider a Granger causality framework.*”

Response: Implemented, and it surfaced and corrected a methodological error of my own. GDELT sentiment is a **weekly** series. An initial pass ran the test at daily frequency by forward-filling the weekly values, which turns the predictor into a step function (constant within each week) that is mechanically incapable of carrying daily lead-lag information — the forward-fill was destroying the signal, and the resulting null was an artefact.

Run correctly at the native weekly frequency, sentiment **Granger-leads** volatility for **10 of 18** asset-sentiment pairs, **7 surviving Benjamini–Hochberg at $q < 0.05$** (12 at $q < 0.10$). This is reported as the one genuinely positive empirical result in the paper, alongside the scope condition, and the daily $\rightarrow$ weekly correction is disclosed candidly.

R2.4 — Diebold-Mariano / out-of-sample forecasting discussion

Comment: “A more thorough discussion ... to compare out-of-sample forecasting performance ... against a standard GARCH(1,1) benchmark.”

Response (candid). As disclosed in the prior round, the out-of-sample DM figures carried in the original draft were not reproducible from the analysis pipeline, which implements in-sample model comparison. I report what the pipeline supports — GJR-GARCH-X attains the lowest AIC for four of six assets — and flag a rolling-window, HLN-corrected out-of-sample DM evaluation as future work. Under the new framing this is lower-stakes: the paper’s contribution is the dependence-robust inference on the event differential, not a forecasting-superiority claim, and I would rather under-claim than report a figure the code does not produce.

R2.5 — Correlation heatmap

Comment: “Consider adding a correlation matrix, perhaps as a heatmap.”

Response: Added (6×6, hierarchically clustered, annotated). The cross-asset correlation it displays (mean $\bar{\rho} \approx 0.69$, range 0.54–0.83) is no longer just descriptive — it is the quantity that drives the entire inference correction (the dependence that makes the six per-asset coefficients pseudoreplicated). The figure therefore does double duty: descriptive in §4.1, and the empirical premise of the four inference lessons.

Table 1 (descriptive statistics) recomputed from the shipped data — transparency/reproducibility correction. Building the heatmap surfaced that the descriptive-statistics table carried numbers from a stale data vintage. We have **recomputed Table 1 from the shipped data so it is now reproducible end to end**: $N = 2,434$ observations per asset (2,191 for BNB, which lists later; 14,361 total); excess kurtosis 9.7–29.4 on raw returns (the fat tails that motivate the Student- t innovations); skewness mild and mixed in sign (–1.05 to +0.51); annualised volatility 52.5% (BTC) to 82.3% (XRP). The earlier round’s “XRP 77.3% as the maximum” was a stale-vintage figure and is corrected to 82.3%. The one inference-critical quantity — the mean pairwise correlation ≈ 0.69 (BTC-ETH 0.83 the strongest pair) — was verified to **reproduce exactly**, so the dependence premise of the whole inference argument is unchanged. We disclose the table refresh candidly because precise descriptive reporting is part of this paper’s pitch.

R2.6 — “Relative independence” wording for $\rho \approx 0.5$

Comment: “A correlation of this magnitude is a moderate-to-strong relationship, not ‘relative independence.’ ”

Response: Rephrased. XRP’s pairwise correlations (0.54–0.65) are described as moderate-to-strong but below the BTC-ETH benchmark. This is consistent with the paper’s central point: even the *least* correlated asset is correlated enough that treating the six as independent is pseudoreplication.

R2.7 — Constant-mean-return assumption made explicit

Comment: “Explicitly mention in Section 3.4 that the models assume a constant average return.”

Response: Done and justified empirically: $r_t = \mu + \varepsilon_t$ with constant μ ; Ljung-Box flags serial correlation in only 8/18 cells and the fitted AR(1) coefficients are tiny. Two *distinct* controls are now reported, and we are careful not to conflate them:

1. **AR(1)-in-mean** — a **serial-correlation control only**. Refitting with an AR(1) mean leaves the event coefficients essentially unchanged (multiplier 5.04× vs 4.88×). This addresses autocorrelation; it does *not* control for event-induced first-moment shocks.
2. **Event dummies in the mean equation** (new this revision) — the control for **event-induced first-moment shocks**. Because the mean equation is otherwise constant-only, any drift on event days is left in the residual, where it can compete with the variance dummy. We re-estimated the curated six-asset panel demeaning each return series by OLS on [const, D_infra, D_reg], so events enter **both** moments. The fitted first-moment event coefficients are small and nearly identical across categories (cross-asset means **−0.45% infrastructure**, **−0.46% regulatory**), and the variance multiplier moves only **4.88×** → **4.42×**. The asymmetry is therefore a second-moment phenomenon, not a relabelled first-moment shock. Shipped as reproducible code (c8i); §*Event dummies in the mean equation*.

Under the framing the point estimate’s stability matters more than its significance, and it is stable under both controls.

R2.8 — Expanded parameter table

Comment: “Expand Table 2 to include α , β , θ_1 , θ_2 , and v .”

Response: Done. A consolidated per-asset parameter table (new, §6.2, Table 4) now reports the full variance-dynamics vector — ω , α , γ , β , v , the persistence $\alpha + \beta + \gamma/2$, and the two event coefficients δ_{infra} , δ_{reg} — for all six assets, with the cross-asset means ($\delta_{\text{infra}} = 1.978$, $\delta_{\text{reg}} = 0.405$) giving the headline 4.88× multiplier; it is reproducible from `code/c14_garch_diagnostics.py`. The two sentiment-decomposition coefficients (θ_{reg} , θ_{infra}) are reported with the Granger analysis (R2.3) rather than duplicated here. The fitted Student- t degrees of freedom ($v \approx 3.1$ – 4.6) are now load-bearing rather than incidental: they are exactly why a Gaussian bootstrap mis-specifies the null (Lesson 3). The small-negative GJR γ estimates are read against Baur and Dimpfl (2018)’s documented inverted crypto leverage effect.

γ (GJR leverage) column correction — carried forward

The data-integrity correction to the γ column disclosed in the prior round is retained: the original positive, starred γ values were stale carry-overs; the correct estimates are the small-negative values (−0.066 to −0.001), sourced directly from the model-output JSON, with the significance stars and the equity-exceeding-leverage claims removed and the negative γ reconciled with the literature (Baur and Dimpfl 2018).

Reproducibility and housekeeping (this revision)

A few non-reviewer changes belong on the record:

- **End-to-end reproducibility.** The full analysis pipeline now runs from a clean clone of the public repository (`github.com/studiofarzulla/crypto-event-study`). We fixed a path-resolution bug, repaired `requirements.txt`, and shipped the previously-missing scripts — the Table-1 generator, a `data_preparation` shim, and the event-in-mean script (c8i). Every number in the manuscript is now reproducible from the repository.

- **Manuscript.** 44 pages, compiles clean. The abstract has been trimmed by roughly 40% to lead on the methodological contribution — treating event inclusion as an explicit, measured design parameter — rather than on the (now-retracted) significance claim.
- **Funding.** The funding statement reads: “No external funding was received.”

Additional pre-submission robustness review (author-initiated)

Because this paper’s whole pitch is inferential discipline, I did not want a referee to be the first to find a hole. Before resubmitting I ran the manuscript and its code through an independent external audit — an adversarial read by a reviewer who had not seen the paper. Nine findings came back. I address each below: the finding as raised, then the honest resolution. Several were already covered by the main revision; two (F1, F3) prompted analyses added this pass; one (F2) is a correct critique that does not bite on this manuscript; the remainder are acknowledged as caveats or fixes.

F1 — Selection on the dependent variable (raised as a potentially fatal flaw). *Finding:* the curated sample may be a high-volatility subset of the candidate pool, so the variance differential is selection on the outcome. **Resolution — refuted on the evidence.** The curation criterion and the mechanical impact screen *disagree* in both directions: of the 50 curated events, 14 fail the ≥ 2 -asset screen yet were retained (4 with *zero* assets moving beyond 1 SD — scheduled regulatory milestones), while 57 screen-passing candidates were dropped, 29 of which moved ≥ 5 of 6 assets. The screen’s pass rate also *favours* regulatory candidates (75.5% vs 64.6%). A high-volatility-subset selection mechanism could produce none of these patterns. Selection here is on event type and salience, not on realised volatility (full detail under R2.1).

F2 — GARCH-X compounding understates peak conditional variance / VaR. *Finding:* a dummy held over a multi-day window does not add a flat δ to variance; it accumulates through the $\beta\sigma^2_{t-1}$ recursion toward $\delta/(1-\beta)$, so a static reading understates the peak conditional variance and any VaR built from it. **Resolution — correct econometrics, but it does not bite here.** The critique is right in general. It does not apply to this manuscript for two reasons. First, the paper contains **no VaR or economic-significance computation** — the static-VaR language the critique targets lived only in the earlier separate version, not here. Second, the amplification is common to both event categories and **cancels in the infrastructure/regulatory ratio**: re-reading the multiplier under full within-window compounding moves it only $4.88\times \rightarrow \approx 5.07\times$, leaving every claim intact. We add a one-line note to the appendix and otherwise the point changes nothing.

F3 — Omitted mean-equation event controls. *Finding:* with a constant-only mean, an event-day first-moment shock is left in the residual and could be relabelled as variance. **Resolution — tested directly.** We re-estimated the curated panel with events in the mean (demeaning on [const, D_infra, D_reg]); the fitted first-moment event effects are small and symmetric (-0.45% infra, -0.46% reg) and the multiplier moves only $4.88\times \rightarrow 4.42\times$. The asymmetry is second-moment (R2.7, code c8i).

F4 — The anticipation test cannot detect diffuse leakage. *Finding:* a single averaged pre-window coefficient rules out a concentrated pre-event volatility spike but cannot exclude low-level information leakage spread diffusely across the pre-window. **Resolution — acknowledged, language softened.** This is correct. The anticipation sweep rules out *concentrated* anticipation (and the multiplier in fact *grows* as the regulatory pre-window lengthens, $4.88\times \rightarrow 8.77\times$), but it cannot exclude diffuse leakage. We have softened the claim from “decisively against anticipation” to “against concentrated anticipation,” and state the diffuse-leakage limitation explicitly.

F5 — No multivariate-GARCH (DCC) discussion. *Finding:* the paper estimates univariate models but never discusses a multivariate-GARCH / DCC alternative for the cross-asset dependence. **Resolution —**

addressed. Cross-asset dependence is not ignored; it is exactly what the Student-*t*-copula CCC-GARCH-X bootstrap — the inference of record — is built to handle, simulating the joint dependence directly rather than approximating its cost through an effective sample size. We now justify the univariate-plus-copula-bootstrap design over a DCC estimator explicitly: the copula bootstrap targets the inferential quantity — the null distribution of the cross-asset coefficient difference under joint tail dependence — directly, without imposing a DCC correlation-dynamics parameterisation we do not separately test. The dependence apparatus is therefore the copula bootstrap, stated rather than left implicit.

F6 — Spurious weekly Granger from temporal aggregation. *Finding:* aggregating a daily series up to weekly can manufacture spurious Granger causality. **Resolution — acknowledged caveat; not the mechanism here.** GDELT sentiment is *natively* weekly, so the test is run at the series’ own frequency rather than aggregated up from daily — the aggregation-induced-spuriousness mechanism does not apply. The surviving pairs are re-confirmed under a Toda–Yamamoto lag-augmented re-test. We nonetheless flag the converse limitation: a weekly-frequency lead cannot be promoted to a daily causal claim, and the manuscript says so (R2.3).

F7 — Zero-imputation of missing sentiment. *Finding:* imputing missing sentiment weeks as zero biases coefficients toward zero. **Resolution — acknowledged as a minor limitation.** Correct; the bias is conservative (toward the null). A logit of the imputed-week indicator on lagged volatility is non-significant for every asset, so the imputation is not tied to volatility, but we note the attenuation as a limitation.

F8 — 5.24× vs 5.7× multiplier mismatch. *Finding:* two curated rolling-winsorisation multipliers (5.24× and 5.7×) are reported and look inconsistent. **Resolution — explained; it is an estimator difference.** Both are curated estimates under the *same* rolling winsorisation; the 5.24× is the FastTARCHX estimator and the 5.7× is the prior paper’s canonical estimator. The gap is an **estimator difference, not a winsorisation difference** (FastTARCHX vs canonical; both event coefficients shift modestly between the two estimators, with δ_{reg} the larger relative mover, $0.419 \rightarrow 0.464$). The present standalone pipeline does not reproduce the 5.7×; the consistent headline throughout is 4.88× (global-clip winsorisation, canonical estimator). The manuscript now states this explicitly (R2.1).

F9 — Code reproducibility (requirements pin, path bug, missing scripts). *Finding:* the public repository did not run end-to-end from a clean clone (a path-resolution bug, a broken `requirements.txt`, and scripts referenced in the text but absent). **Resolution — fixed.** We repaired the path-resolution bug and `requirements.txt`, shipped the Table-1 generator and a `data_preparation` shim, and added the event-in-mean script. The full pipeline now runs end-to-end from a clean clone of github.com/studiofarzulla/crypto-event-study.

F10 — Effective sample size: the contrast, not the coefficient level. *Finding:* the abstract and Lesson 2 led with an effective sample size of ≈ 1.4 , computed from the cross-asset correlation of the per-asset coefficient *levels* ($\bar{\rho} \approx 0.69$); but the significance test is on the infrastructure–regulatory *contrast*, which differences out the common market component. **Resolution — corrected to the estimand-relevant object.** The contrast’s cross-asset correlation is $\rho_d = 0.23$ (DEFF 1.70–2.16), giving an effective $N \approx 2.8$ — a number the manuscript already derived in the Lesson-1 footnote but did not carry into the headline. The abstract and Lesson 2 now lead with the contrast’s effective N (≈ 3 , not the level’s 1.4) and the design-effect rung’s $p \approx 0.07$ – 0.15 , consistent with the body. This is an honesty correction in the *conservative* direction: it stops the result from reading as “barely one observation” when the estimand-relevant penalty is milder. The inference of record (copula bootstrap, $p \approx 0.32$) is unaffected — it never uses an effective- N approximation.

F11 — The size study is a self-consistency check, not a proof of correctness under the true DGP. *Finding:* the Monte-Carlo size table’s *t*-copula calibration recovers nominal size partly *by construction* — the copula is calibrated under the same multivariate-*t* DGP it is then tested against — so the “self-validation” heading

claimed more authority than a calibration check earns. *Resolution — rescoped explicitly.* The passage now states that the check certifies (a) the simulator is correctly implemented and (b) the procedure is correctly sized under a DGP matched to the data’s salient features (heavy Student-*t* margins at $v \approx 3$, the estimated cross-asset correlation) — but *not* under the true, unknown DGP, since the modelling assumptions have been moved into the generator. Rather than leave robustness to that as an assumption, this revision runs it directly (new Table, §5.4, `code/c10b_dgp_perturbation.py`): the size loop is re-estimated under lighter tails ($v=8$), heavier tails ($v=2.5$), a near-Gaussian dependence structure, and a deliberately *mis-specified* calibration (simulate $v=2.5$, calibrate at the fitted v). Findings: (a) the **naive rule over-rejects at 0.48–0.59 in every scenario** — its failure is DGP-independent, so the central diagnostic does not rest on a flattering DGP; (b) the *t*-copula controls size under correctly specified generators (0.03–0.06 across matched rows); (c) candidly, the *t*-copula is *not* immune to misspecification — when the true tails are heavier than its calibration it over-rejects (0.29). That last point cuts in the conclusion’s favour: an *anti-conservative* (over-eager) test that still returns $p = 0.322$ on the data makes the failure to reject more secure, not less. The size study’s load-bearing result — that the naive and Gaussian-copula alternatives over-reject — is independent of the calibration and now shown DGP-robust.

F12 — Scope-condition language tightened; ratio reported as a derived quantity; tone rebalanced.

Finding: three smaller slips, all in the direction of the prose running slightly ahead of the statistics. (a) The drop-out census shows the *mechanical screen* does not favour infrastructure, but §7’s relaxed-threshold subsection let that “support” the mechanism reading against a *curation-bias* alternative the census explicitly cannot adjudicate. (b) The multiplier was treated as the headline object despite a near-zero regulatory denominator that makes it volatile (the anticipation sweep drives it to $8.77\times$ purely by shrinking δ_{reg}). (c) The dominant “artefact / dual null” register understated a coherent directional signal. *Resolution.* (a) §7 now subordinates the census conclusion to its own caveat — it makes the impact-threshold concern unlikely without ruling out a subtler curation bias. (b) The conclusion now flags the ratio as a volatile summary and names the difference $\delta_{\text{infra}} - \delta_{\text{reg}}$ in squared-% points as the steadier underlying object. (c) A synthesis sentence — *robust to inference, fragile to identification, significant under neither* — and a rebalanced close report the directional reading (uniform sign across all six assets and both moments, mechanism-consistent concentration) as **unresolved, not absent**.

I am glad I ran this audit before you did. The findings that required new work — F1’s concrete screen-vs-curation evidence and F3’s event-in-mean check — both strengthen the paper; F8/F9 close the two places a careful referee would have caught an inconsistency; and F10–F12 are the precision pass that aligns the headline framing with the (already-correct) body, correcting the manuscript’s one remaining places where the rhetoric had run ahead of the statistics — including, notably, in the direction of *overstating* the null.

On the title and scope change

The original title asserted an “asymmetric volatility response.” The revised title — *Do Cryptocurrency Markets Differentiate Infrastructure from Regulatory Shocks? A Multi-Moment Event Study with Dependence-Robust Inference* — poses the question the data can honestly address and signals both the multi-moment design and the inference focus. The paper now subsumes the companion first-moment (returns) analysis, so the question is answered once, at both moments, on one sample, under one design. I recognise that retracting my own headline against an open Major-Revision slot is unusual; I have made the change because correct inference required it, and I have written the manuscript to make the editors’ judgment on re-review straightforward.

Summary of changes

Area	Change	Origin
Title / framing	Retitled to the multi-moment question; dual failure-to-reject is now the conclusion; inference cautionary tale is the lead contribution	R2.1 → full inference reckoning
Headline	Prior $5.7\times / p < 0.001$ retracted; present pipeline's reproducible curated point estimate is 4.88× (global-clip, canonical estimator), directional-but-non-significant at both moments (returns $p = 0.283$; variance t -copula $p = 0.322$; contrast eff $N \approx 2.8$, design-effect $p \approx 0.07\text{--}0.15$)	Defects 1–2
New first-moment analysis	CAR comparison (+7.19 pp) via event-level block bootstrap + Ibragimov–Müller; underpowered caveat (MDE $\approx 19\text{--}20$ pp); subsumes the <i>no-structure</i> companion	new + R1.1
New inference section	Four lessons: pseudoreplication, concentrating dependence, heavy-tailed bootstrap innovations, block vs parametric	Defects 1–2
New: inference ladder	Single table walking the identical comparison through 7 inference rungs: naive 0.0008, model-free proxies ≈ 0.5 , t -copula 0.322 — rungs bracket, not monotone	Defects 1–2
New: Monte-Carlo size study	True-null DGP; naive rule over-rejects 51% @ 5% ($\approx 10\times$ nominal); t -copula bootstrap restores nominal size (3.9%); self-validating	new (c10)
Inference of record (variance)	Student- t -copula CCC-GARCH-X bootstrap ($B = 2000$); replaces the Gaussian draw; Gaussian-vs- t comparison table across all 3 break specs	Defect 2

Area	Change	Origin
§Scope condition	Curated 4.88× (headline) vs mechanical 1.3–1.6× (never significant); concrete selection evidence (14/50 retained despite failing screen — 4 with zero assets; 57 dropped, 29 moved $\geq$ 5/6); drop-out census (reg passes higher, 75.5% vs 64.6%); 208-pool figure removed, 135 the screening pool	R2.1
Estimator bookkeeping	4.88× = headline (global-clip, canonical); 5.24× = FastTARCHX (rolling); 5.7× = prior canonical (rolling), not reproduced here; 5.24-vs-5.7 = estimator difference in δ_{reg} (0.464 vs 0.419), not winsorisation	R2.1, F8
New: event dummies in mean	Curated panel demeaned on [const, D_infra, D_reg]; first-moment effects small/symmetric (−0.45% / −0.46%); multiplier 4.88× $\rightarrow$ 4.42×; second-moment, not relabelled first-moment shock	R2.7, F3 (c8i)
Expanded control battery	Break-regime table (robust p), anticipation sweep table (4.88× $\rightarrow$ 8.77×), winsorisation invariance, AR(1) serial-correlation control + event-in-mean control, persistence SEs	R2.2
§Granger	Weekly sentiment-leads-volatility (10/18; 7 at FDR $q < 0.05$) with full per-asset table; daily forward-fill artefact disclosed	R2.3
Bai-Perron	Breaks at FTX; within-regime persistence ≈ 0.905 (descriptive only, 0/27 drops > 2 SE); crisis-control robust inference reported	R2.2, R1 minor
Fuller appendices	t -copula bootstrap algorithm (step-by-step), GARCH-X spec, Monte-Carlo DGP	new

Area	Change	Origin
Presentation	Heatmap (R2.5), full parameter table (R2.8), constant-mean justification (R2.7), “relative independence” rephrased (R2.6), network demoted (R1.4), exogeneity explicit (R1.2), overlap protocol (R1 minor)	R1.x, R2.x
Corrections carried forward	γ column, units (variance not volatility), AIC 4/6, retracted Corbet (2019) reference removed	prior round
DM / out-of-sample	In-sample AIC reported; out-of-sample DM flagged as future work; unreproducible figures removed	R2.4
Table 1 recomputed	Regenerated from shipped data (was stale vintage): $N = 2,434 / 2,191$ BNB / 14,361 total, excess kurtosis 9.7–29.4, skew -1.05 to $+0.51$, ann. vol 52.5–82.3% (XRP max, not 77.3%); $\bar{\rho} \approx 0.69$ / BTC-ETH 0.83 reproduce exactly	R2.5
Reproducibility	Pipeline runs end-to-end from clean clone (path-bug fix, requirements.txt repair, Table-1 generator + data_preparation shim + c8i shipped); 44pp; abstract trimmed $\approx 40\%$; funding “No external funding received”	this revision
New: pre-submission audit	Author-initiated external code+manuscript review, F1–F12 addressed point-by-point	this revision
Precision pass (F10–F12)	Effective- N corrected to the contrast (≈ 2.8 , not the level’s 1.4); size study rescoped as a self-consistency check; census subordinated to its caveat; ratio flagged as derived (difference is the steadier object); tone rebalanced to “unresolved, not absent” with the <i>robust-to-inference</i> / <i>fragile-to-identification</i> / <i>significant-under-neither</i> synthesis	this revision (final read)

The paper is a more honest object than the original submission. The selection-bias comment that prompted this revision did exactly what good refereeing should: it pushed me to inspect the inference, and the inference did not hold. I have followed it where it led.